Optimal Parameters for Maximum Achievable Efficiency

Companion Parameter Tables for Figure: `eta_vs_alpha_all_cases.pdf`

Anharmonic Quantum Otto Engine Study

July 22, 2026

Overview

This document provides the complete set of optimized parameter values $(\beta_c, \beta_h, \omega_c, \omega_h)$, maximum achievable efficiency $\eta_{\max}$, and corresponding work output W_{ext} for each configuration (Case 1 to Case 4) as a function of the anharmonicity parameter $\alpha \in [0.00, 0.20]$. These parameter values correspond directly to the curves plotted in `eta_vs_alpha_all_cases.pdf`.

Physical Constraints Enforced

All optimizations enforce strict thermodynamic and platform boundaries:

1. **Platform Ratio Caps:** $\frac{\omega_h}{\omega_c} \leq 7.0$ and $\frac{\beta_c}{\beta_h} \leq 25.0$ (based on diamond NV-centre platform caps).
2. **Perturbation Validity:** $\varepsilon_c \leq 0.10$ and $\varepsilon_h \leq 0.10$, where $\varepsilon = \frac{3\alpha \coth(\beta\omega/2)}{4\omega^3}$.
3. **Engine Operation Condition:** $\beta_c\omega_c > \beta_h\omega_h$ and $W_{\text{ext}} > 0$.

1. Case 1: Symmetric Coupling ($\lambda_{ab} = \lambda_{cd} = 1$)

Table 1: Optimized parameters for Case 1 ($\lambda_{ab} = \lambda_{cd} = 1$).

Anharmonicity α	β_c	β_h	ω_c	ω_h	**Max Efficiency** $\eta_{\max}$	**Work** W_{ext}
0.00	9.1223	0.6338	1.5789	11.0524	**0.857143**	0.008601
0.05	9.1223	0.6338	1.5789	11.0524	**0.856057**	0.008573
0.10	9.1223	0.6338	1.5789	11.0524	**0.854972**	0.008544
0.15	9.1223	0.6338	1.5789	11.0524	**0.853886**	0.008516
0.20	9.1223	0.6338	1.5789	11.0524	**0.852800**	0.008488

2. Case 2: Asymmetric Hot Stroke ($\lambda_{ab} = f(\omega), \lambda_{cd} = 1$)

3. Case 3: Asymmetric Cold Stroke ($\lambda_{ab} = 1, \lambda_{cd} = f(\omega)$)

4. Case 4: Dual Asymmetric Strokes ($\lambda_{ab} = f(\omega), \lambda_{cd} = f(\omega)$)

Table 2: Optimized parameters for Case 2 ($\lambda_{ab} = f(\omega), \lambda_{cd} = 1$).

Anharmonicity α	β_c	β_h	ω_c	ω_h	**Max Efficiency** $\eta_{\max}$	**Work** W_{ext}
0.00	2.5000	0.1000	1.0000	3.6001	**0.629889**	3.775108
0.05	2.5000	0.1000	1.3106	4.4104	**0.565322**	3.292993
0.10	2.5000	0.1000	1.5396	4.8929	**0.537954**	3.101196
0.15	2.5000	0.1000	1.7008	5.2009	**0.519848**	2.975815
0.20	2.5000	0.1000	1.8298	5.4329	**0.506088**	2.880982

Table 3: Optimized parameters for Case 3 ($\lambda_{ab} = 1, \lambda_{cd} = f(\omega)$).

Anharmonicity α	β_c	β_h	ω_c	ω_h	**Max Efficiency** $\eta_{\max}$	**Work** W_{ext}
0.00	2.5000	0.1000	1.0000	3.0568	**0.397207**	3.287285
0.05	2.5000	0.1000	1.7900	4.8024	**0.349186**	2.705745
0.10	2.5000	0.1000	2.1002	5.3378	**0.335188**	2.533113
0.15	2.5000	0.1000	2.3134	5.6805	**0.326132**	2.423390
0.20	2.5000	0.1000	2.4810	5.9389	**0.319325**	2.341971

Table 4: Optimized parameters for Case 4 ($\lambda_{ab} = \lambda_{cd} = f(\omega)$).

Anharmonicity α	β_c	β_h	ω_c	ω_h	**Max Efficiency** $\eta_{\max}$	**Work** W_{ext}
0.00	2.5000	0.1000	1.0000	2.2581	**0.343051**	2.828437
0.05	2.5000	0.1000	1.8673	3.7290	**0.284554**	2.214085
0.10	2.5000	0.1000	2.1961	4.2017	**0.269055**	2.048489
0.15	2.5000	0.1000	2.4212	4.5102	**0.259288**	1.945551
0.20	2.5000	0.1000	2.5983	4.7458	**0.252073**	1.870182